

$$\rho_X = \text{Diagram 1} = \text{Diagram 2}$$

The diagram illustrates the decomposition of a quantum state ρ_X into a sequence of three components:

- Diagram 1:** A red rounded rectangle labeled Ψ^* is connected to another red rounded rectangle labeled Ψ . The input and output lines are connected in a loop, indicating a trace operation.
- Diagram 2:** A sequence of three components connected in series: a yellow circle labeled U^* , a white circle labeled λ^2 , and a yellow circle labeled U .